

Hematopoiesis & the Lymphoid System

Shubham Adroja, MD • built to follow alongside the 62-slide lecture deck

Official Week 7 LOs

Slide-order companion

3 ScholarRx Bricks

Histology recognition

Lymph + RBC circulation maps

Factoid sheet companion

Overview: Official Week 7 Learning Objectives

LO 1. Describe embryonic development of the blood system and the sites of child and adult hematopoiesis.

LO 2. Understand bone marrow hematopoiesis and the concept of hematopoietic stem cells.

LO 3. Recognize the difference between T cells and B cells and know their sites of origin and maturation in bone marrow and thymus.

LO 4. Recognize normal histology of lymph nodes, tonsil, gut-associated lymphoid tissue, spleen, and thymus.

LO 5. Describe the anatomic location and biologic function of the major lymphoid structures: lymph node chains, Peyer's patches, tonsils/adenoids, spleen, and thymus.

LO 6. Describe circulation of cells, blood, and lymphatic fluid through lymph nodes and circulation of red blood cells through the spleen.

SCHOLARRX BRICKS PAIRED WITH THIS LECTURE

Erythropoiesis

Hematopoiesis: Myeloid & Lymphoid Lineages

Anatomy of the Lymphatic System

LO 1

Slides 3-8: yolk sac -> AGM -> liver/spleen -> marrow, HbF/HbA switching, and where hematopoiesis lives from birth through adulthood.

LO 2

Slides 9-19: marrow compartments, HSC properties/markers, lineage tree, erythroid/granulocyte/megakaryocyte maturation, and normal marrow recognition.

LO 3

Slides 20-28 + 55-60: B cells mature in marrow; T cells originate in marrow but mature in thymus through positive/negative selection.

LO 4-6

Slides 29-60: lymph node, MALT/tonsil, spleen, thymus histology plus lymphocyte, lymph-fluid, and splenic RBC circulation.

SOURCE HIERARCHY

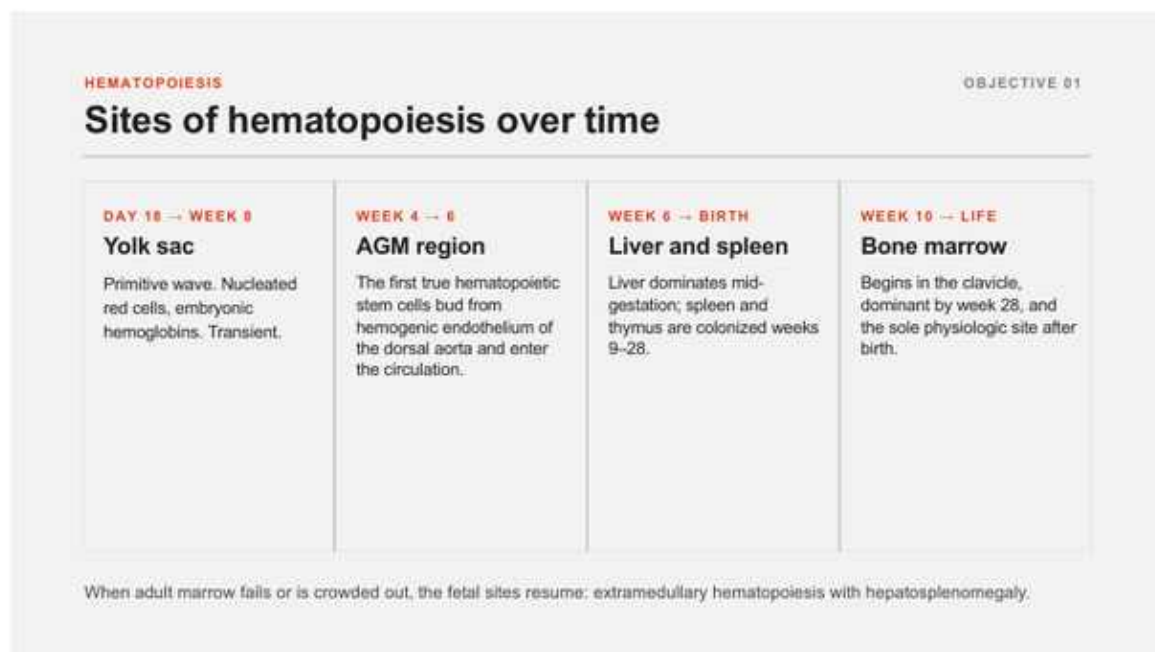
The **62-slide Adroja deck** is the primary source and controls slide order, terminology, and timing. The uploaded companion is used only as supplementary board-style reinforcement. One important example: the companion uses the older simplified “yolk sac -> liver -> spleen -> marrow” timeline, while the current deck explicitly includes the **AGM region** and starts marrow hematopoiesis earlier. The companion is not allowed to overwrite the current lecture.

THE WHOLE LECTURE IN ONE SENTENCE

Blood cells are generated in changing developmental sites, then adult marrow maintains every lineage; lymphocytes are educated in primary organs and encounter antigen in secondary organs whose architecture is built around efficient cell and antigen traffic.

Slide 4: primitive hematopoiesis begins with blood islands

- At about **day 18-19**, yolk-sac blood islands generate primitive blood and primitive vasculature together.
- Outer cells become endothelium; inner cells become blood.
- This wave is transient and is dominated by **nucleated erythroid cells with embryonic hemoglobins** plus primitive macrophages.



Lecture slide 5 - full slide retained. This is the timing slide to learn from the current deck: yolk sac first, definitive HSC emergence from the AGM region, liver/spleen expansion, then marrow becoming dominant before birth and remaining the physiologic site thereafter.

DEVELOPMENTAL TIMELINE - CURRENT LECTURE

Day 18 -> week 8: yolk sac, primitive wave.

Week 4-6: AGM region, first true HSCs from hemogenic endothelium.

Week 6 -> birth: liver dominates; spleen/thymus are colonized.

Week 10 -> life: bone marrow begins and becomes dominant by about week 28.

CLINICAL CORRELATE

When adult marrow cannot meet demand or is crowded out, fetal sites can reactivate
-> **extramedullary hematopoiesis with hepatosplenomegaly.**

Slides 6-7: HbF and hemoglobin switching

- HbF = **alpha2 gamma2**; it has higher oxygen affinity than HbA and supports placental oxygen transfer.
- The lecture places HbF at roughly 70% at birth and under 2% by one year, while HbA becomes dominant after birth.
- **Alpha chains are needed prenatally**, so severe alpha-thalassemia can present in utero.
- **Beta-chain disorders emerge after the postnatal switch**, explaining delayed presentation of severe beta-thalassemia and sickle-cell disease.
- The physiologic hemoglobin nadir at about 8-12 weeks reflects falling EPO drive after birth plus this transition.

Sites of child and adult hematopoiesis

Birth	All cavities are red marrow, including hands and feet.	
Childhood	Progressive fatty conversion from the distal extremities inward. Tibial aspiration remains valid in infants.	
Adult	Vertebrae, sternum, ribs, pelvis, skull, and the proximal epiphyses of femur and humerus. Sampled at the posterior superior iliac spine.	
Under stress	Yellow marrow reconverts to red in chronic hemolysis, high altitude, or growth-factor therapy.	

Lecture slide 8 - full slide retained. At birth essentially all marrow cavities are red. During childhood fatty conversion proceeds from distal extremities inward. Adult hematopoiesis is concentrated in axial bones plus proximal humerus/femur; stressed yellow marrow can reconvert.

ADULT RED MARROW

Vertebrae • sternum • ribs • pelvis • skull • proximal humerus/femur.

Posterior superior iliac spine = standard adult sampling site.

Architecture of the marrow cavity

COMPARTMENT 1

Endosteum

The bone-lining surface with its osteoblasts. Quiescent stem cells are held here, out of cycle.

COMPARTMENT 2

Hematopoietic cords

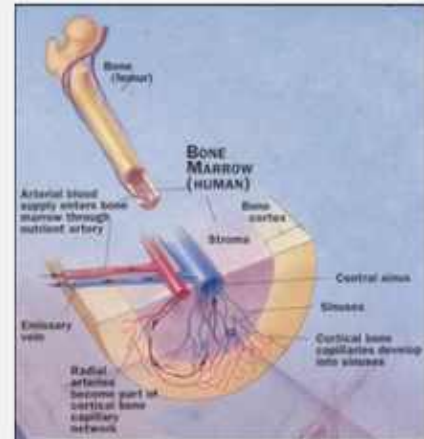
Developing cells in a stromal meshwork. Erythroblastic islands cluster around a central macrophage; megakaryocytes about the sinusoid wall.

COMPARTMENT 3

Sinusoids

Thin-walled vessels draining to the central vein. Mature cells cross the endothelium to enter the circulation.

Normal myeloid-to-erythroid ratio in an adult aspirate: 2:1 to 4:1.

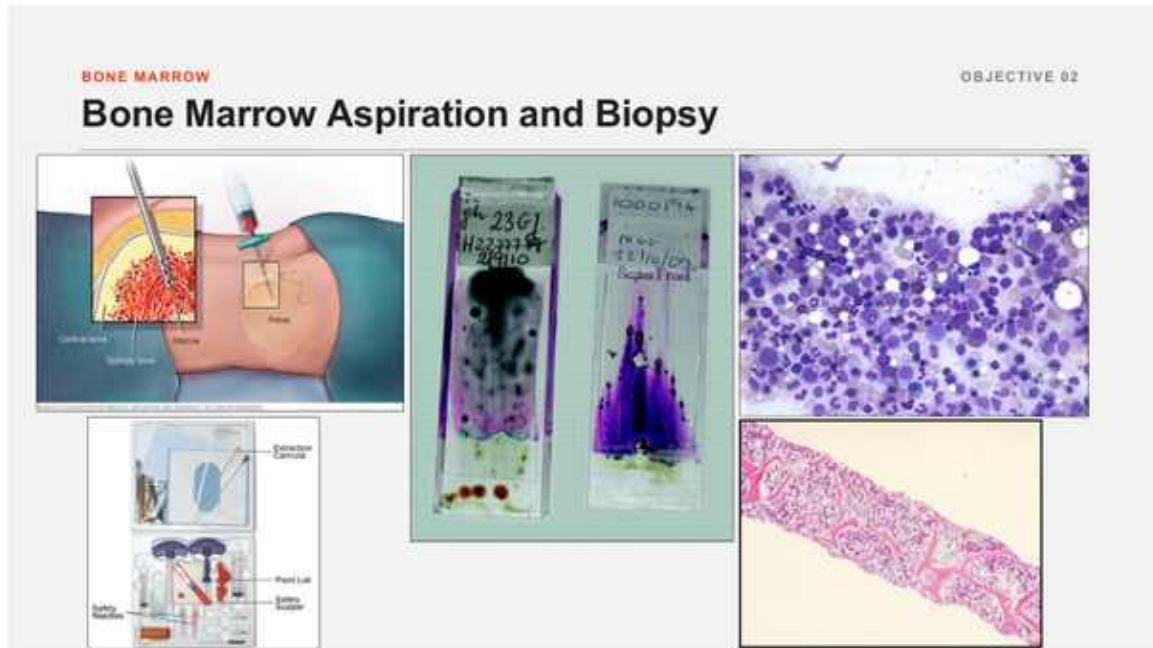


Lecture slide 9 - full slide retained. Adroja divides marrow into the endosteal surface, hematopoietic cords, and sinusoids. The spatial arrangement explains stem-cell quiescence, erythroblastic islands, megakaryocyte position, and release of mature cells.

Compartment	What lives there / why it matters
Endosteum	Bone-lining surface; quiescent HSC niche.
Hematopoietic cords	Developing cells in stroma; erythroblastic islands around central macrophages.
Sinusoids	Thin-walled vessels; mature cells cross into blood. Megakaryocytes sit against the sinus wall.

NORMAL ADULT MARROW RATIO

Myeloid : erythroid = about 2:1 to 4:1.



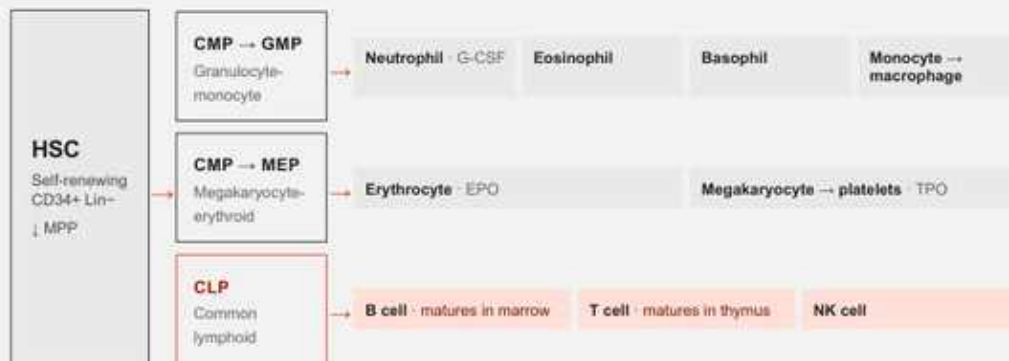
Lecture slide 10 - full slide retained. Aspiration gives a cell-rich smear for morphology/differential; core biopsy preserves architecture and cellularity. They answer overlapping but different questions.

	Aspirate	Core biopsy
Best for	Individual-cell morphology, lineage percentages, flow/cytogenetic samples	Architecture, cellularity, fibrosis, focal infiltrates
Think	"What cells are here?"	"How is the marrow built?"

Slide 11: hematopoietic stem-cell properties

- **Self-renewal:** preserves the stem-cell pool.
- **Multipotency:** one HSC can reconstitute myeloid and lymphoid lineages.
- **Quiescence:** most HSCs are out of cycle, limiting mutation and protecting them from cycle-active chemotherapy.
- **Rare:** roughly 1 in 10,000-100,000 nucleated marrow cells in the lecture.
- Lecture immunophenotype: **CD34+ CD38- Lin-.**
- G-CSF can mobilize HSCs into peripheral blood for apheresis collection.

The hematopoietic lineage tree



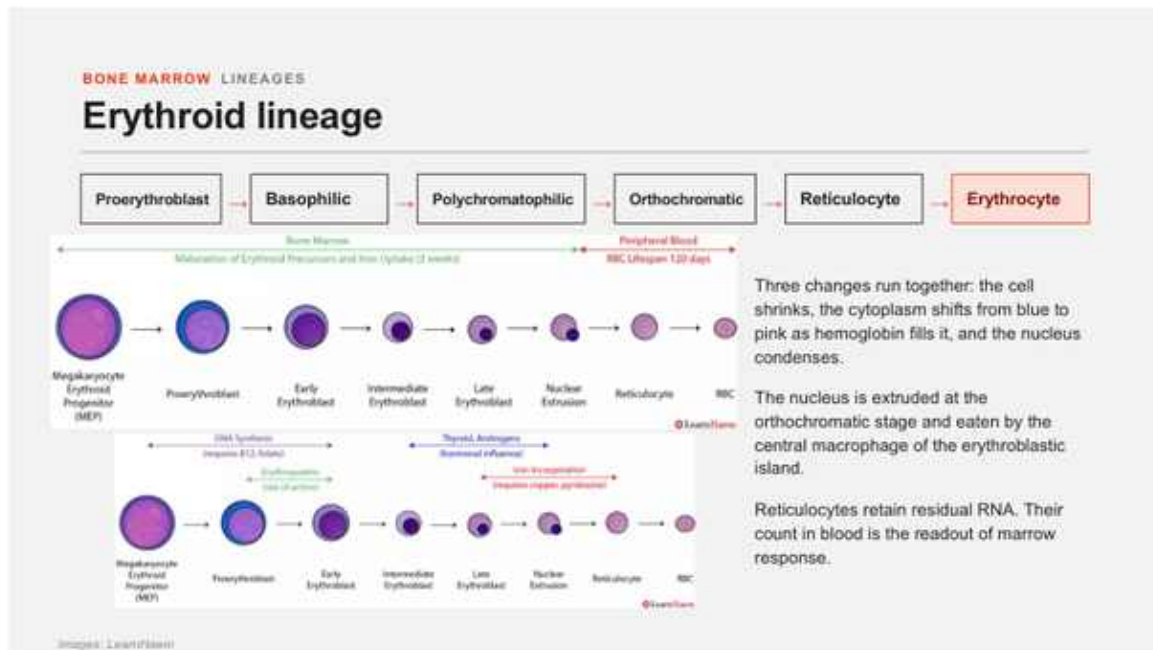
Older nomenclature you will still meet: CFU-GEMM for the multipotent myeloid progenitor, BFU-E and CFU-E for the erythroid steps, named for the colonies they form in culture.

Lecture slide 12 - full slide retained. The lineage map separates common myeloid and common lymphoid output and anchors the major growth factors: EPO for erythroid cells, TPO for megakaryocytes/platelets, and G-CSF for neutrophil production.

GROWTH-FACTOR ANCHORS

EPO -> RBC. TPO -> platelets. G-CSF -> neutrophils.

Erythroid maturation



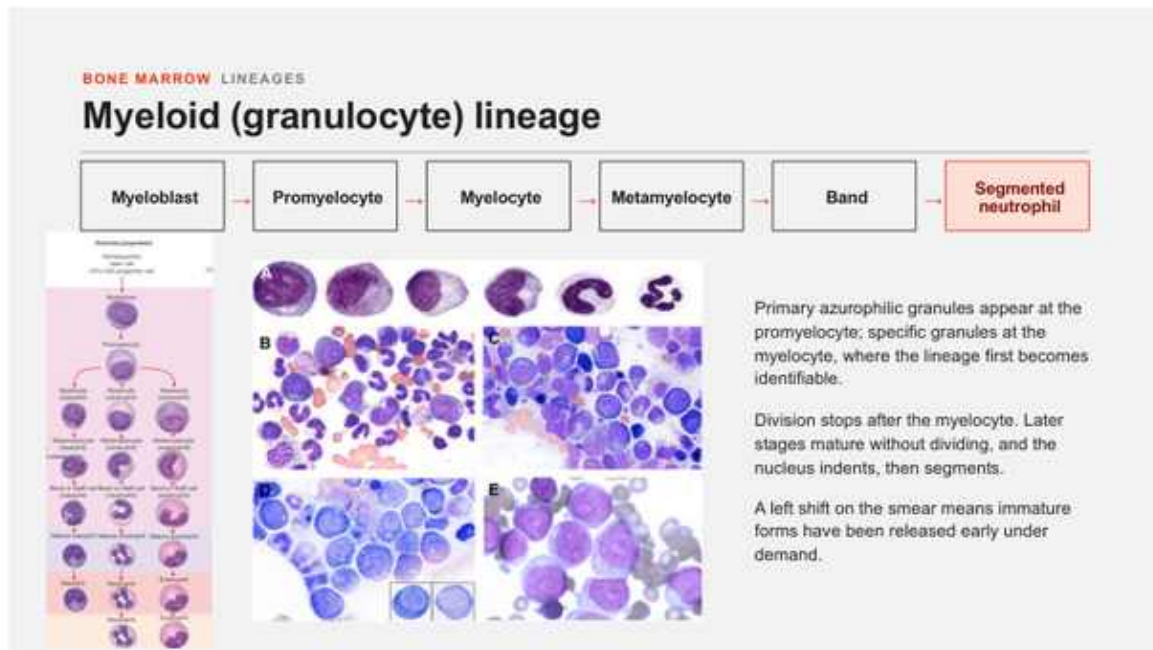
Lecture slide 14 - full slide retained. As erythroid cells mature, cell size falls, cytoplasm shifts blue to pink as hemoglobin accumulates, and the nucleus condenses before extrusion at the orthochromatic stage.

ERYTHROID SEQUENCE

Proerythroblast -> basophilic -> polychromatophilic -> orthochromatic -> reticulocyte -> erythrocyte.

Reticulocytes retain residual RNA, so the retic count is a readout of marrow response.

Granulocyte maturation



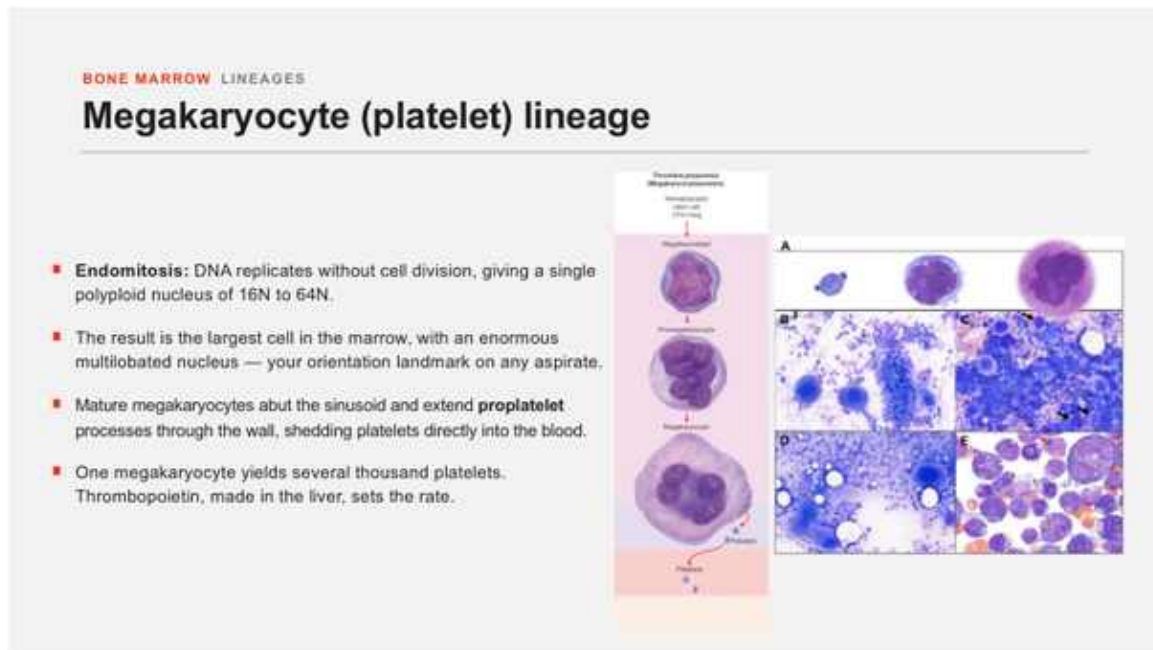
Lecture slide 15 - full slide retained. Primary azurophilic granules appear at the promyelocyte stage; specific granules at the myelocyte stage make lineage recognition easier. Division stops after the myelocyte.

NEUTROPHIL SEQUENCE

Myeloblast -> promyelocyte -> myelocyte -> metamyelocyte -> band -> segmented neutrophil.

A left shift means immature forms are released into blood early under demand.

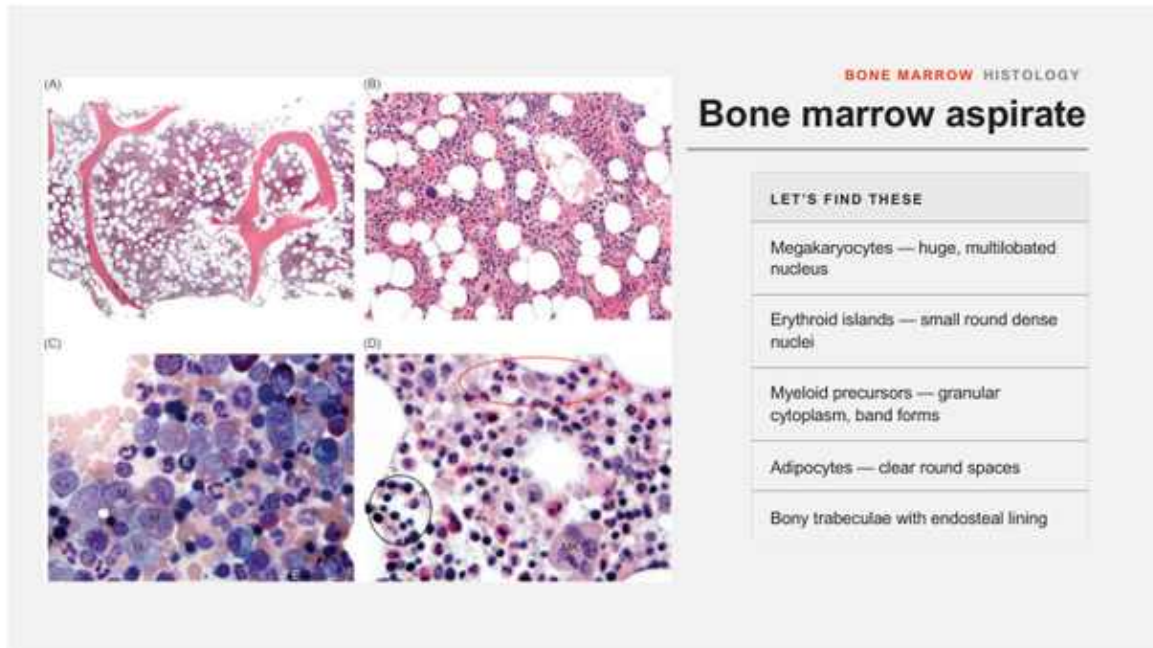
Megakaryocytes and platelets



Lecture slide 16 - full slide retained. Megakaryocytes undergo endomitosis: DNA replication without cell division. Their huge multilobulated polyploid nucleus makes them a useful marrow landmark.

Megakaryocyte logic

- Polyploid nucleus: lecture gives approximately **16N-64N**.
- Mature cells abut marrow sinusoids.
- Proplatelet processes extend through the sinusoidal wall and shed thousands of platelets.
- **TPO is produced by the liver** and regulates platelet production.



Lecture slide 17 - full slide retained. The slide asks you to orient yourself using megakaryocytes, erythroid islands, granular myeloid precursors, adipocytes, and bony trabeculae.

Marrow feature	Recognition clue
Megakaryocyte	Huge cell with multilobulated nucleus
Erythroid island	Cluster of small erythroid precursors around a macrophage
Myeloid precursors	Granular cytoplasm; maturation spectrum including bands
Adipocyte	Clear round space in routine histology
Bony trabecula	Hard eosinophilic bone framework with endosteal lining

LO 2 checkpoint

Can you draw the HSC tree, name the HSC properties/markers, identify the three marrow compartments, and recognize each major lineage by maturation pattern?

LYMPHOID SYSTEM OBJECTIVE 05

Primary and secondary lymphoid organs

PRIMARY - GENERATIVE

Bone marrow · Thymus

Lymphocytes are produced and educated here. The repertoire is generated and self-reactive clones are removed. Antigen-independent.

SECONDARY - PERIPHERAL

Lymph nodes · Spleen · Tonsils · Peyer's patches · MALT

Where naive lymphocytes meet antigen, expand clonally, and differentiate into effector and memory cells. Antigen-dependent.

Lecture slide 21 - full slide retained. Primary/generative organs create and educate the repertoire without antigen-driven activation. Secondary/peripheral organs are where naive cells encounter antigen and expand.

	Organs	Main job
Primary / generative	Bone marrow, thymus	Production, maturation, repertoire generation, removal of strongly self-reactive clones
Secondary / peripheral	Lymph nodes, spleen, tonsils, Peyer's patches, MALT	Antigen encounter -> clonal expansion -> effector + memory cells

Lymphoid lineage: T and B cells

T-CELL LINEAGE

Marrow → thymus

Progenitors leave the marrow and mature in the thymus, where positive and negative selection shape the TCR repertoire. Mature T cells enter the circulation.

B-CELL & NK LINEAGE

Marrow → periphery

B cells rearrange immunoglobulin genes and are screened against self within the marrow, then leave as naive cells. NK cells share the common lymphoid progenitor but need no thymic step.

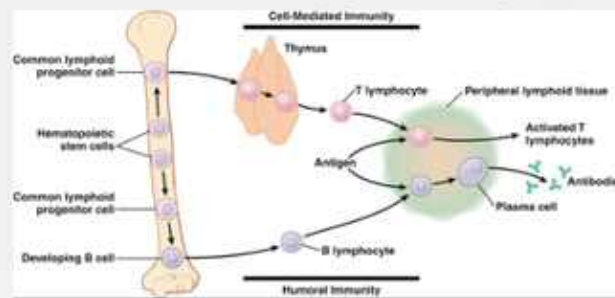
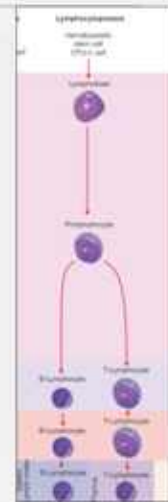


Image: Guyton and Hall Textbook of Medical Physiology 14e, Chapter 35



Lecture slide 22 - full slide retained. Both originate from marrow progenitors. B cells complete maturation in marrow; T-cell precursors leave marrow and mature in thymus. NK cells share a common lymphoid progenitor but do not need a thymic step.

THE EXAM POINT OF LO 3

B cell: born + matures in marrow.

T cell: born in marrow, matures in thymus.

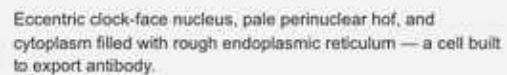
Slides 23-24: B vs T cell function

- B-cell sequence: pro-B → pre-B → immature B → mature naive B; heavy chain first, light chain second, then surface IgM.
- Strong self-reactivity in marrow triggers receptor editing or deletion: central B-cell tolerance.
- T cells use a TCR and recognize **processed peptide presented by another cell**, not free native antigen.
- Helper T cells coordinate responses; cytotoxic T cells kill infected targets.

Lymphocyte and plasma cell histology



References



Dendritic cells sample peripheral tissues, mature after danger signals, travel through **afferent lymphatics**, and present antigen to naive T cells in the **paracortex** of the draining node. This route becomes important again when we map lymph-node architecture.

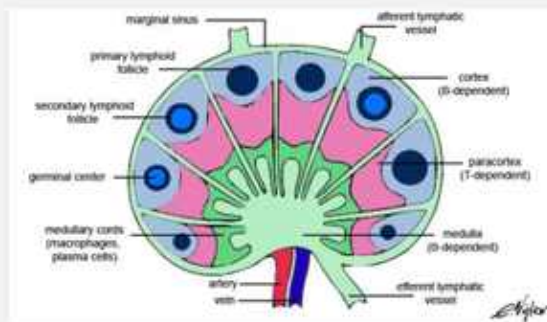
Why nodes exist

Nodes concentrate **naïve B cells, naïve T cells, antigen, and antigen-presenting cells** in the same place. They also filter lymph through macrophage-lined sinuses before fluid returns to blood.

Slide 32: major node chains

- **Cervical:** head/neck and Waldeyer's ring.
- **Axillary:** upper limb, breast, chest wall.
- **Mediastinal/hilar:** lung, heart, esophagus.
- **Mesenteric:** bowel.
- **Para-aortic:** kidney, gonad, retroperitoneum.
- **Iliac:** pelvic viscera.
- **Inguinal:** lower limb/perineum/external genitalia.

Lymph node histology



OUTSIDE IN

Dense collagenous **capsule**, with trabeculae

Subcapsular sinus — where afferent lymph arrives

Cortex — B cell follicles, with or without germinal centers

Paracortex — T cell zone, contains high endothelial venules

Medulla — cords rich in plasma cells, sinuses rich in macrophages and reticular cells

Hilum — artery, vein, efferent lymphatic

Lecture slide 33 - full slide retained. The slide provides the core histology map: capsule -> subcapsular sinus -> cortex/B follicles -> paracortex/T zone with HEVs -> medulla/cords and sinuses -> hilum.

LYMPH NODE OUTSIDE -> IN

Capsule -> subcapsular sinus -> cortex (B) -> paracortex (T + HEV) -> medulla (cords + sinuses) -> hilum.

Zone	Dominant cells / landmark
Cortex	B-cell follicles; secondary follicles have pale germinal centers
Paracortex	T-cell zone + high endothelial venules
Medullary cords	Plasma cells, lymphocytes, macrophages
Medullary sinuses	Lymph channels rich in macrophages/reticular cells

BOARD REINFORCEMENT FROM THE COMPANION

In severe T-cell deficiency such as DiGeorge syndrome, the **paracortex** is depleted because it is the T-cell zone.

Slide 36: lymph fluid itself

- Interstitial hydrostatic pressure generates lymph.
- Muscle contraction helps propel it through one-way lymphatic vessels.
- Regional chains progressively collect lymph.
- Return to blood occurs through the **thoracic duct** and **right lymphatic duct**.
- Clinical correlate: lymph-node removal can interrupt drainage and cause chronic lymphedema.

LYMPH NODES OBJECTIVE 06

Two routes into the node

FROM THE LYMPHATICS

Antigen and dendritic cells

Afferent vessels pierce the convex capsule and empty into the subcapsular sinus. Soluble antigen reaches B cells; dendritic cells migrate on into the paracortex.

FROM THE BLOODSTREAM

Resting T and B cells

Naïve lymphocytes leave the blood by crossing high endothelial venules in the paracortex, then sort themselves into their zones.

ONE EXIT Everything leaves by the efferent lymphatic at the hilum — on to the next node in the chain, and finally back to the bloodstream via the thoracic duct or right lymphatic duct.

HEVs are found in the paracortex of lymph nodes and in mucosal lymphoid tissue — but not in the spleen, which has no afferent lymphatics and admits cells directly from the blood.

Nature Reviews | Immunology

Homing and cellular traffic in lymph nodes. Nat Rev Immunol 3, 557-570 (2003)

Lecture slide 37 - full slide retained. Antigen and dendritic cells arrive through afferent lymphatics into the subcapsular sinus; resting B and T cells arrive from blood through HEVs in the paracortex. Everything exits through the efferent lymphatic at the hilum.

TWO ROUTES IN, ONE ROUTE OUT

Lymphatics IN: antigen + dendritic cells -> subcapsular sinus.

Blood IN: naive T/B cells -> HEV in paracortex.

OUT: one efferent lymphatic at the hilum.

Ozzy checkpoint - click him after you can trace both routes without looking.

HEV DISTINCTION

HEVs are in lymph-node paracortex and mucosal lymphoid tissue. **The spleen does not use HEVs for the same entry route because the spleen has no afferent lymphatics and samples blood directly.**

Slide 38: lymphocyte recirculation

Resting cells repeatedly cycle **blood -> HEV -> node -> efferent lymph -> thoracic duct -> blood**. After activation they spend days proliferating, change adhesion/homing programs, then leave for inflamed tissues.

LO 4-6 node checkpoint

Can you label cortex/paracortex/medulla, put B/T/plasma cells in the right place, and separately trace **antigen/DC traffic** versus **resting lymphocyte traffic**?

Slides 39-44

MALT, Peyer's patches, appendix, and tonsils

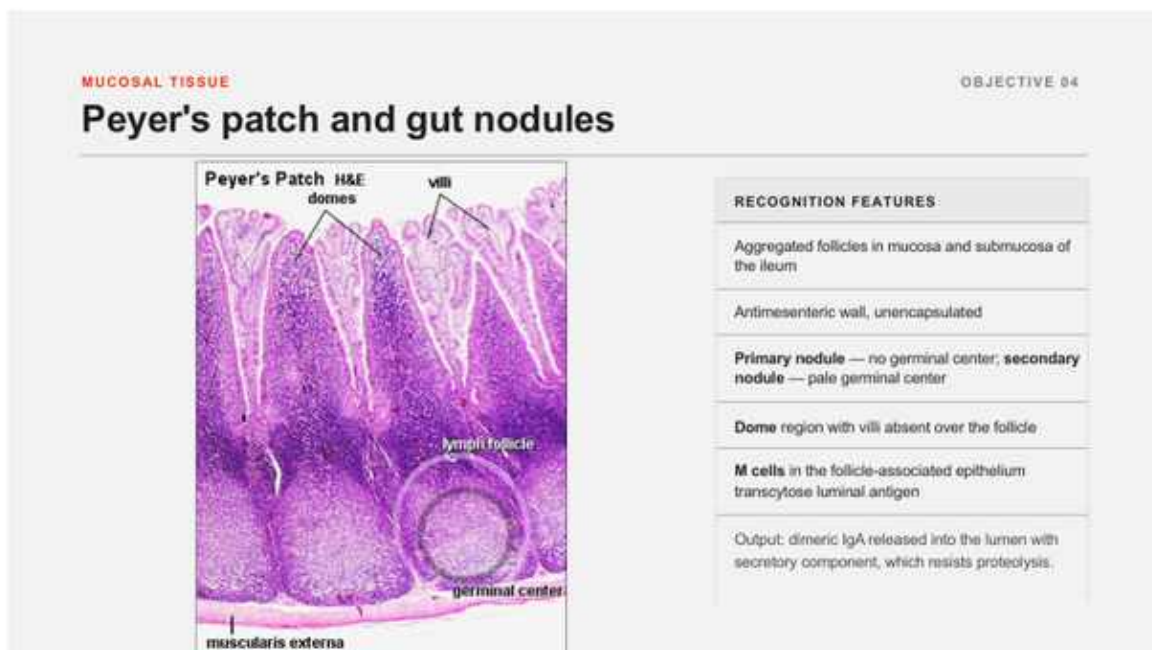
LO 4-5

MUCOSAL-ORGAN PRINCIPLE

Unlike lymph nodes, mucosal lymphoid tissue is generally **unencapsulated and positioned immediately beneath an epithelium**. Antigen can cross the epithelial surface directly, so these sites do not need afferent lymphatics.

Slide 40: the mucosal structures

- **Diffuse lymphoid tissue:** scattered lymphocytes/plasma cells throughout the lamina propria.
- **Waldeyer's ring:** palatine, pharyngeal/adenoid, and lingual tonsils.
- **Peyer's patches:** aggregated follicles concentrated in the ileum.
- **Appendix:** circumferential lymphoid follicles surrounding the lumen.



Lecture slide 41 - full slide retained. Peyer's patches are unencapsulated aggregated follicles in ileal mucosa/submucosa on the antimesenteric side. M cells transcytose luminal antigen across follicle-associated epithelium.

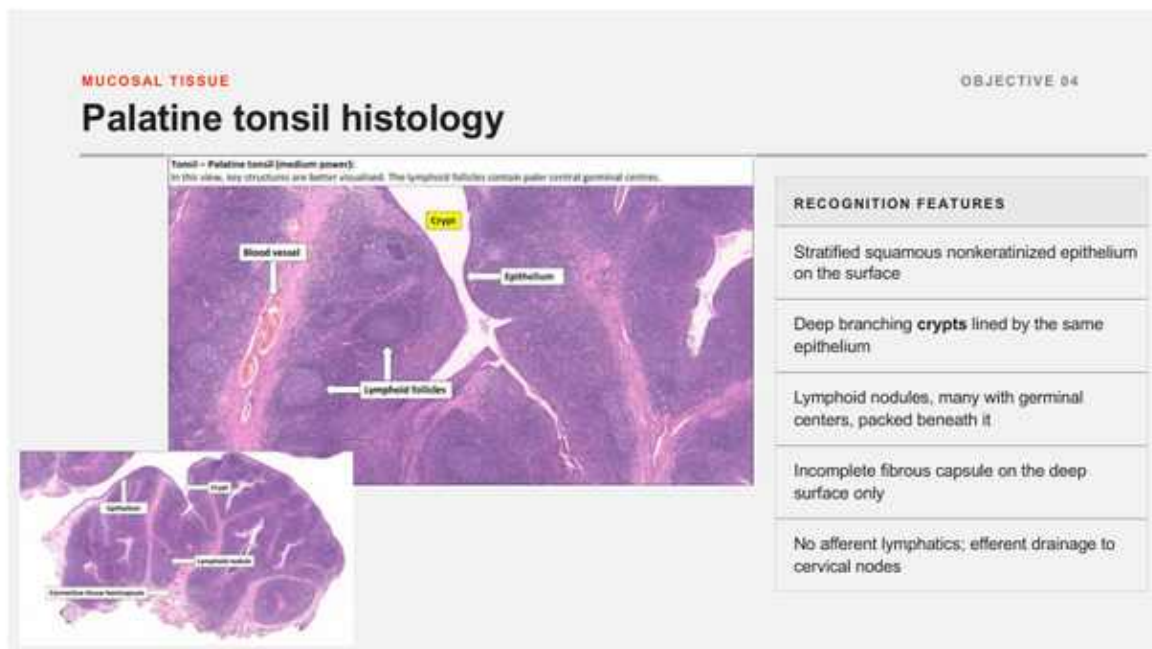
PEYER'S PATCH RECOGNITION

Ileum + aggregated follicles + absent villi over dome + M cells.

Mucosal B-cell output supports **dimeric secretory IgA**.

Slides 42-43: appendix and Waldeyer's ring

- **Appendix:** narrow irregular lumen, colonic crypts, no villi, complete ring of lymphoid follicles.
- **Palatine tonsil:** paired, between tonsillar pillars, squamous epithelium with deep crypts.
- **Pharyngeal tonsil/adenoid:** roof of nasopharynx, respiratory epithelium, folds rather than deep crypts.
- **Lingual tonsil:** base of tongue.



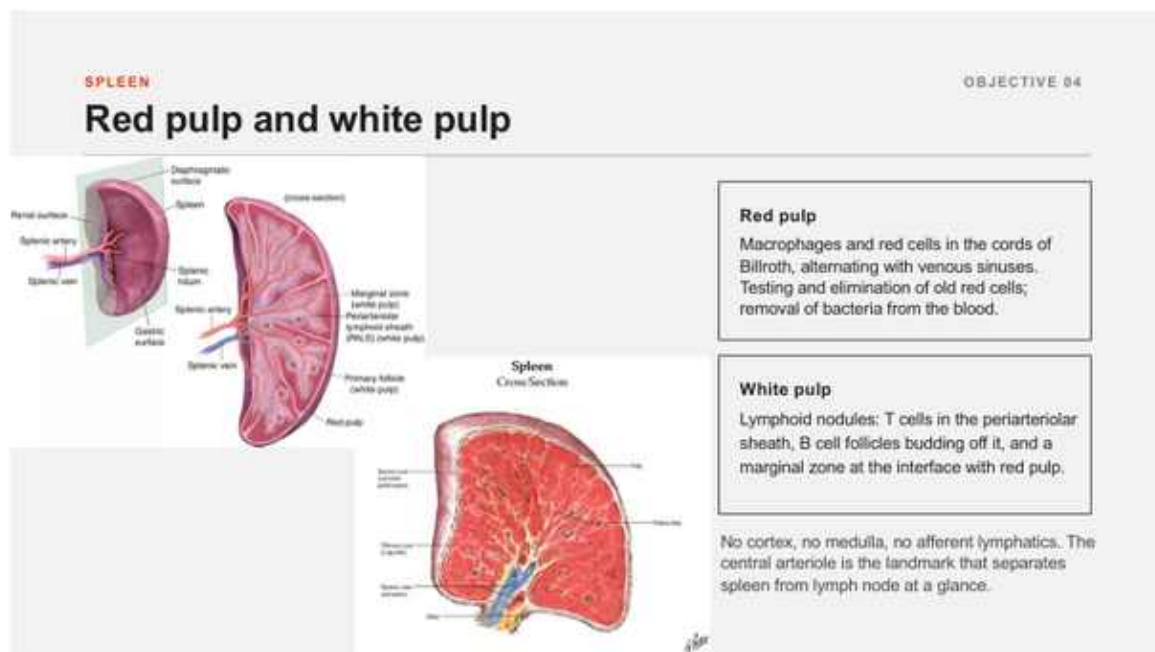
Lecture slide 44 - full slide retained. The palatine tonsil is identified by nonkeratinized stratified squamous epithelium, deep branching crypts, underlying lymphoid nodules, and an incomplete capsule on the deep surface.

PALATINE TONSIL

Squamous epithelium + deep crypts + lymphoid follicles.

Slide 46: location and jobs

- Left hypochondrium, deep to ribs 9-11; intraperitoneal.
- Filters **blood**, not lymph.
- Removes blood-borne antigen/immune complexes and encapsulated bacteria.
- Tests aged/abnormal RBC deformability and recycles iron.
- Stores a platelet reservoir and can resume extramedullary hematopoiesis under stress.

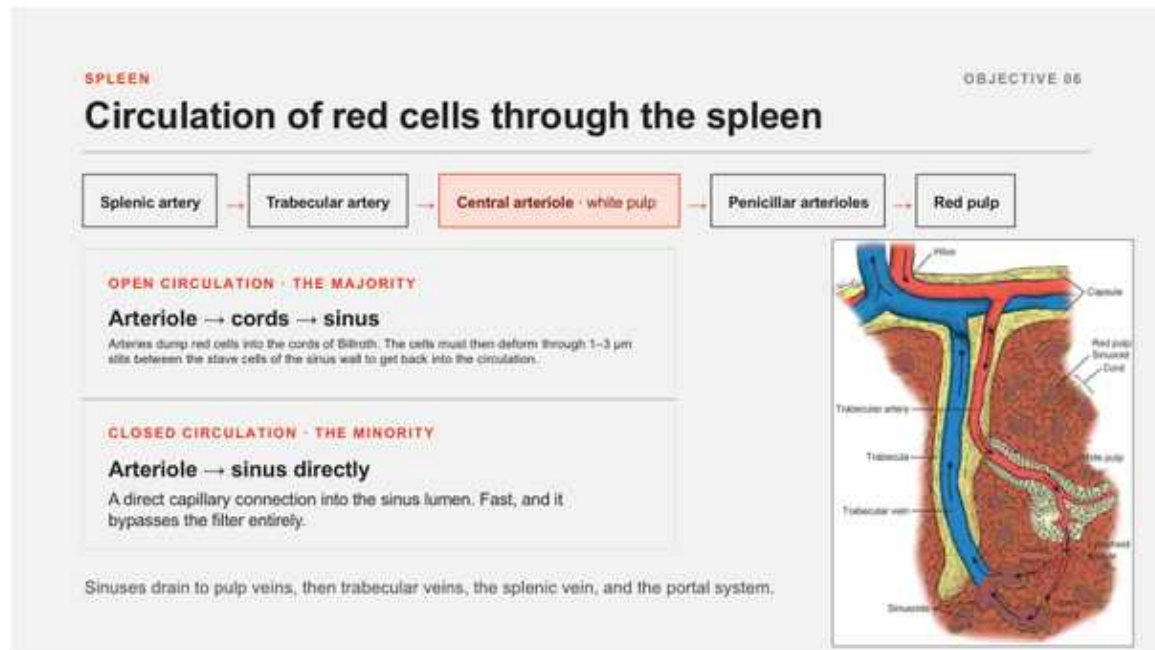


Lecture slide 47 - full slide retained. Red pulp contains cords of Billroth and venous sinuses for RBC filtration. White pulp contains T-cell PALS around central arterioles, B-cell follicles, and a marginal zone.

Spleen zone	Core function / landmark
Red pulp	RBC-rich cords + sinusoids; mechanical filtration and macrophage clearance
PALS	T-cell zone wrapped around central arteriole
White-pulp follicle	B-cell zone
Marginal zone	Interface with red pulp; macrophages + memory B cells see blood-borne antigen first

HISTOLOGY DIFFERENTIATOR

Spleen has no cortex, no medulla, and no afferent lymphatics. A central arteriole embedded in white pulp is the visual landmark that separates spleen from lymph node.



Lecture slide 50 - full slide retained. Most splenic RBC flow is open circulation: arterioles empty into cords of Billroth, and RBCs must deform through tiny slits between sinus stave cells to return to blood. Stiff cells fail this test and are removed.

SPLENIC RBC ROUTE

Splenic artery -> trabecular artery -> central arteriole/white pulp -> penicillar arterioles -> red pulp -> cords -> sinus -> splenic vein.

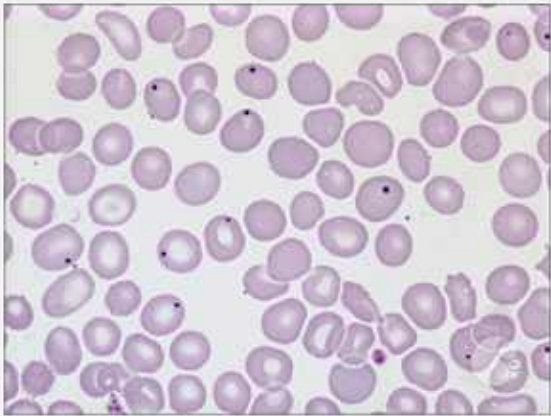
Open circulation = majority and performs the deformability test.

Slides 51-54: what happens without or with too much spleen

- Splenectomy can reduce immune destruction in refractory ITP/AIHA and hemolysis in selected membrane/enzyme disorders.
- After splenectomy, abnormal cells persist: **Howell-Jolly bodies, target cells, acanthocytes**.
- The same smear pattern with a spleen present suggests **functional asplenia**, classically sickle-cell disease.
- Vaccination against encapsulated organisms is essential before elective splenectomy; fever afterward is an emergency.
- Splenomegaly can result from infiltration, portal congestion, chronic hemolytic workload, or extramedullary hematopoiesis.

SPLEEN CLINICAL

The blood smear after splenectomy



Abnormal and aged red cells persist in the circulation because nothing is culling them.

Howell-Jolly bodies — residual nuclear fragments — survive because nothing is pitting them out. Target cells and acanthocytes appear alongside them.

The same picture in a patient who still has a spleen means functional asplenia, as in sickle cell disease.

Image: ASH image bank

Lecture slide 52 - full slide retained. Howell-Jolly bodies are residual nuclear fragments normally removed by splenic pitting; they persist when splenic function is absent.

BOARD REINFORCEMENT FROM THE COMPANION

Asplenia classically increases risk from **encapsulated organisms**, especially pneumococcus, Haemophilus influenzae type b, and meningococcus. Pair that clinical risk with the morphologic clue of Howell-Jolly bodies.

Spleen checkpoint

Can you say **blood filter**, map red vs white pulp, identify PALS, trace open circulation, and explain both Howell-Jolly bodies and post-splenectomy infection risk?

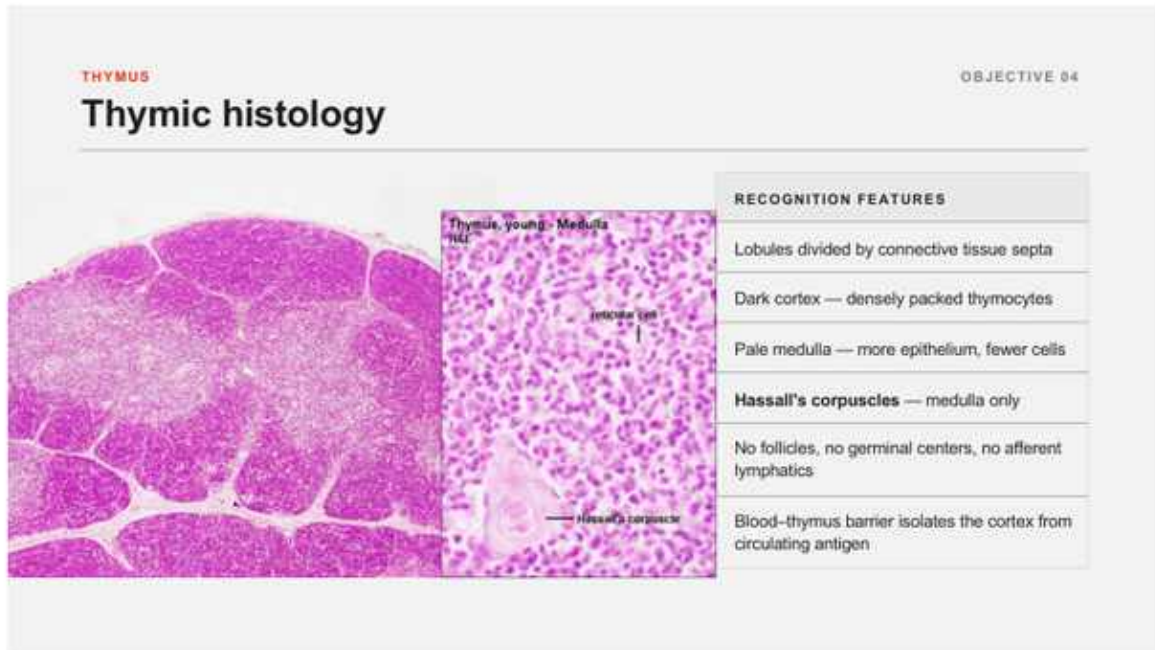
Slides 55-60

Thymus: T-cell education, histology, and involution

LO 3-5

Slides 56-57: location and embryology

- Anterior mediastinum behind the manubrium, over great vessels/pericardium.
- Thymic **epithelium derives from the third pharyngeal pouch**; lymphocytes themselves migrate in from marrow.
- DiGeorge syndrome links failed pouch development with thymic aplasia/T-cell deficiency, hypoparathyroidism, and conotruncal cardiac defects.



Lecture slide 58 - full slide retained. Thymic lobules have a dark cortex and pale medulla. Hassall corpuscles occur only in the medulla. There are no lymphoid follicles or germinal centers.

THYMUS RECOGNITION

Lobules + dark cortex + pale medulla + Hassall corpuscles.

No follicles. No germinal centers. No afferent lymphatics.

T cell maturation in the thymus

Arrival

T cell precursors from the bone marrow enter at the corticomedullary junction as double-negative thymocytes.

Positive selection

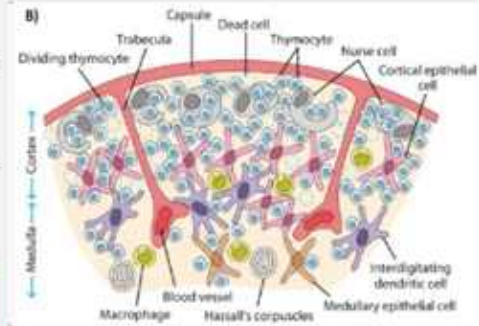
Double-positive cells are tested on cortical epithelium for the ability to engage self-MHC. No engagement means death by neglect. MHC restriction is set here.

Negative selection

Dendritic cells and medullary epithelium present self-peptide; cells that bind too strongly are deleted. AIRE supplies tissue-restricted antigens.

Exit

Single-positive CD4 or CD8 cells leave the medulla by post-capillary venules into the peripheral circulation.



~98% of thymocytes die in the thymus. The cost of a repertoire that is both useful and self-tolerant.

Lecture slide 59 - full slide retained. T-cell precursors arrive from marrow, undergo positive selection in cortex for self-MHC recognition, then negative selection in medulla against strong self-reactivity. Survivors exit as single-positive CD4 or CD8 cells.

Stage	Where	Question being asked
Arrival	Corticomedullary junction	Bone-marrow precursor enters as immature thymocyte
Positive selection	Cortex	Can the TCR interact with self-MHC at all?
Negative selection	Medulla	Does it bind self too strongly? AIRE helps display tissue-restricted antigens.
Exit	Medulla -> circulation	Single-positive CD4 or CD8 mature naive T cell

SELECTION MNEMONIC

Positive = cortex = keep cells that can see MHC.

Negative = medulla = delete cells that attack self too strongly.

Slide 60: involution

The thymus reaches greatest absolute weight around puberty and then becomes progressively replaced by adipose tissue. New T-cell output falls with age, but residual thymic epithelium can regenerate a repertoire after marrow transplantation.

Synthesis

Histology recognition and traffic map

LO 4-6

Structure	T-cell zone	B-cell zone	Instant recognition clue
Lymph node	Paracortex	Cortical follicles	Capsule + cortex/paracortex/medulla + subcapsular sinus + HEV
Spleen	PALS	White-pulp follicle	Central arteriole + red/white pulp; no cortex/medulla
Thymus	Almost entirely T lineage	None as follicles	Lobules, dark cortex, pale medulla, Hassall corpuscles
Palatine tonsil	Interfollicular	Follicles	Squamous epithelium + deep crypts
Peyer's patch	Interfollicular	Aggregated follicles	Ileal mucosa/submucosa + dome/M cells + no capsule
Appendix	Interfollicular	Ring of follicles	Narrow lumen + colonic crypts + no villi + circumferential lymphoid tissue

FILTER QUESTION

Lymph node filters lymph.

Spleen filters blood.

Thymus educates T cells.

Bone marrow produces all blood cells and matures B cells.

THREE CIRCULATION PATHWAYS WORTH DRAWING ONCE

Node antigen/DC: tissue -> afferent lymphatic -> subcapsular sinus -> B-cell cortex / DC to paracortex -> efferent lymph.

Node resting lymphocyte: blood -> HEV -> node -> efferent lymph -> thoracic duct -> blood.

Spleen RBC: splenic arterial tree -> red-pulp cords -> squeeze through sinus slits -> venous return.

Learning-Objective Synthesis

Official LO	What you should be able to do without notes
LO 1	Give the current lecture's yolk sac -> AGM -> liver/spleen -> marrow timeline and identify adult active-marrow sites.
LO 2	Explain HSC self-renewal/multipotency/quiescence, CD34+ CD38- Lin-, marrow compartments, lineages, and key growth factors.
LO 3	State B vs T origin/maturation and explain cortical positive versus medullary negative selection in thymus.
LO 4	Recognize lymph node, palatine tonsil, Peyer's patch/appendix, spleen, and thymus from their architecture.
LO 5	Match each major lymphoid structure to location and biologic function.
LO 6	Trace both routes into a lymph node, the single efferent route out, lymphocyte recirculation, and splenic RBC open circulation.

LAST 60 SECONDS

Embryo: yolk sac -> AGM -> liver/spleen -> marrow.

HSC: CD34+ CD38- Lin-; self-renewing + multipotent + quiescent.

B: matures marrow. **T:** matures thymus.

Node: cortex B, paracortex T/HEV, medulla plasma cells/sinuses.

MALT: unencapsulated beneath epithelium; Peyer's = ileum/M cells; palatine tonsil = squamous crypts.

Spleen: blood filter; PALS T; follicle B; red pulp tests RBC deformability.

Thymus: dark cortex/pale medulla/Hassall; positive cortex, negative medulla.

Source hierarchy: Adroja's 62-slide Hematopoiesis & Lymphoid System deck is the primary source and the official Week 7 objective sheet defines emphasis. The uploaded companion is supplementary and is used only for board-style reinforcement where it agrees with the current deck; conflicting simplified developmental timing is not substituted for the current lecture.